EE 5239 Nonlinear Optimization: Stochastic Gradient Descent (a)

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Announcement

- We will start discussing different “topics” in optimization and applications

Outline

More on the SVM example

- We worked through the SVM example and saw why transforming to dual problem is useful

- One key point: after minimizing the Lagrangian dual function, we have

$$\mathbf{x} = \sum_i \lambda_i b_i \mathbf{a}_i$$

- So when use the CD, each time pick a single λ_j^{r+1} to update

$$\begin{aligned}\mathbf{x}^{r+1} &= \sum_i \lambda_i^{r+1} b_i \mathbf{a}_i = \sum_i \lambda_i^{r} b_i \mathbf{a}_i + (\lambda_j^{r+1} - \lambda_j^r) b_j \mathbf{a}_j \\ &= \mathbf{x}^r + (\lambda_j^{r+1} - \lambda_j^r) b_j \mathbf{a}_j\end{aligned}$$

- Compare with perceptron learning algorithm (j is a miss classified point)

$$\mathbf{x}^{r+1} = \mathbf{x}^r + b_j \mathbf{a}_j$$

Outline

SVM Again...

- Again let's look at the SVM problem
- The problem is given by (non-separable case)

$$\begin{aligned} \min \quad & \frac{1}{2}\boldsymbol{x}^T\boldsymbol{x} + c\sum_i \xi_i \\ \text{subject to} \quad & b_i(\mathbf{a}_i^T\boldsymbol{x}) \geq 1 - \xi_i,\ \xi_i \geq 0\ \forall\ i. \end{aligned} \tag{0.1}$$

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- Let's take an alternative view ...

$$\min_{\boldsymbol{x}} \quad \frac{1}{N}\underbrace{\sum_{i=1}^{N} \max\{0, 1 - b_i(\mathbf{a}_i^T\boldsymbol{x})\}}_{\text{hinge loss}} + \gamma\boldsymbol{x}^T\boldsymbol{x} \tag{0.2}$$

- Equivalent formulation?

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- An equivalent formulation (with $\gamma = \frac{1}{2cN}$)

- The first term, hinge loss, average loss over the data sets

- The second term, regularization

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SVM Again...

- **First observation.** The problem becomes unconstrained
- **Second observation.** The first term represents the empirical loss, similar to other models
- **Third observation**. the first "loss" term is *nonsmooth* (at some points, there could be no gradient available)!
- All the algorithms we learned don't work
- In practice the problem can be huge! For example a dataset used by KDD Cup 2012 contains the advertisement data from certain search engine, which has 641,707 distinct advertisement items with N = 235,582,879 data samples [1]
- What if we further want to process one (or a subset, a.k.a. mini-batch of) data set at a time?

Illustration

- SVM on a text data corpus (An example by Leon Bottou)
- Reuters RCV1 document corpus
  1. Predict a category of a document
  2. $N = 781,000$ training examples (documents)
  3. $K = 50,000$ features (one feature per word)

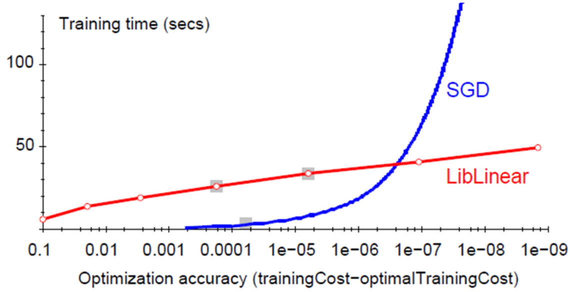

Figure 0.1: Performance of Stochastic SVM (SGD SVM), by Leon Bottou.

General Story

- SVM is just an example, generally, many optimization problems can be formulated by

$$\min_{\boldsymbol{x}} \quad \underbrace{\sum_{i=1}^{N} \ell_i(\boldsymbol{x}; \mathbf{a}_i, b_i)}_{\text{empirial loss on training}} \quad + \quad \underbrace{r(\boldsymbol{x})}_{\text{regularization}} \tag{0.4}$$

- Optimize the data one by one? Handle nonsmoothness in the objective?

- Primal dual approach as we did in SVM doesn't always work, needs a general way to deal with this situation

- To deal with the nonsmooth function, need a new notion "subgradient"; To handle multiple terms, "stochastic" gradient method (SGD)

Outline

More on Convexity

- Recall that we learned that if a function $f(x)$ is *twice differentiable*, then it is convex over a set X iff

$$\nabla^2 f(x) \succeq 0, \; \forall \; x \in X$$

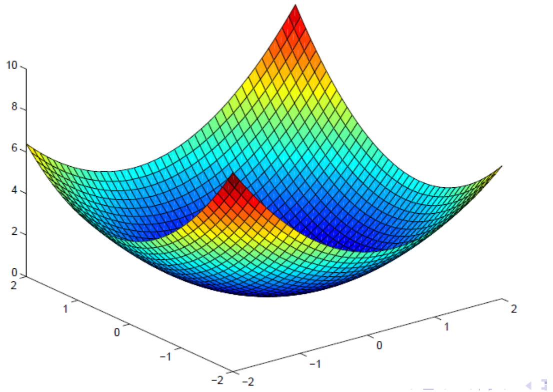

Figure 0.1: Illustration of a convex quadratic function.

More on Convexity

- What if a function is non-differentiable? Still convex?
- How does the function $\max\{0, 1-x\}$ look like? Convex? Why?
- **The subgradient**, $\partial f(x)$ of f at x is a set that

$$\partial f(x) := \{g \mid f(y) \geq f(x) + \langle g, y - x\rangle, \forall\ y\}$$

- **Claim**: IF $\partial f(x)$ has a single element at x, then $f(x)$ is differentiable at x

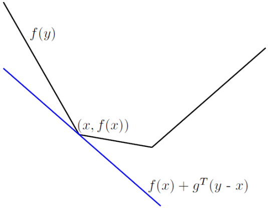

Figure 0.2: A function that is convex but non-differentiable.

Exercise

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$$|x|, \quad \max\{0, 1 - x\}$$

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- In particular, for the hinge loss $h(\boldsymbol{x}) = \max\{0, 1 - b_i \mathbf{a}_i^T \boldsymbol{x}\}$, the following is one subgradient

$$\partial h(\boldsymbol{x}) = -b_i \mathbf{a}, \text{ if } b_i(\mathbf{a}_i^T \boldsymbol{x}) < 1, \quad \partial h(\boldsymbol{x}) = 0, \text{ Otherwise}$$

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- The subgradient of two functions
$\partial(h(\boldsymbol{x}) + g(\boldsymbol{x})) = \partial h(\boldsymbol{x}) + \partial g(\boldsymbol{x})$

Subgradient Method

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

- $f(x)$ is convex, but nonsmooth
- Subgradient Method

$$x^{r+1} = x^r - \alpha^r g^r$$

where α^r is a stepsize, $g^r \in \partial f(x)$ is one of the subgradient

- **Note**: This is NOT a descent method!

Optimality

- Recall for f convex, differentiable,

$$f(x^\star) = \inf_x f(x) \Leftrightarrow 0 = \nabla f(x^\star)$$

- Generalization to nondifferentiable convex f:

$$f(x^\star) = \inf_x f(x) \Leftrightarrow 0 \in \partial f(x^\star)$$

proof. by definition (!)

$$f(y) \geq f(x^\star) + 0^T(y - x^\star) \text{ for all } y \Leftrightarrow 0 \in \partial f(x^\star)$$

Subgradient Method: Convergence

- Main proof steps a bit different than gradient descent
- Assume that the subgradients are all bounded, $\|g^r\| \le G$
- We have

$$\begin{aligned}\|x^{r+1} - x^*\|^2 &= \|x^r - \alpha g^r - x^*\|^2 \\ &= \|x^r - x^*\|^2 - 2\alpha\langle g^r, x^r - x^*\rangle + \alpha^2\|g^r\|^2 \quad (0.1)\end{aligned}$$

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- Utilizing the definition of subgradient

$$f(x^*) \ge f(x^r) + \langle g^r, x^* - x^r\rangle = f(x^r) - \langle g^r, x^r - x^*\rangle \quad (0.2)$$

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- We obtain

$$\begin{aligned}&\|x^{r+1} - x^*\|^2 \le \|x^r - x^*\|^2 + 2\alpha[f(x^*) - f(x^r)] + \alpha^2\|g^r\|^2 \\ \Rightarrow &f(x^r) - f(x^*) \le \frac{\|x^r - x^*\|^2 - \|x^{r+1} - x^*\|^2}{2\alpha} + \frac{\alpha}{2}G^2\end{aligned}$$

Subgradient Method: Convergence

- Summing over $t = 1 \to T$, we have

$$
\begin{aligned}
&\sum_{t=1}^{T} f(x^r) - f(x^*) \\
&\le \frac{1}{2\alpha} \sum_{r=1}^{T} \left(\|x^r - x^*\|^2 - \|x^{r+1} - x^*\|^2 \right) + \frac{T\alpha G^2}{2} \\
&= \frac{1}{2\alpha} \|x^1 - x^*\|^2 - \frac{1}{2\alpha} \|x^{T+1} - x^*\|^2 + \frac{T\alpha}{2} G^2
\end{aligned}
$$

Subgradient Method: Convergence

- Now let $D = \|x^1 - x^*\|$, which is a constant, we have

$$f(x^{\text{best}}) - f(x^*) \leq \frac{1}{T}\sum_{t=1}^{T}(f(x^r) - f(x^*)) \leq \frac{1}{2\alpha T}D^2 + \frac{\alpha}{2}G^2$$

$f(x^{\text{best}})$ is the best objective function we have so far (up until T iterations)

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  $f(x^{\text{best}})$ is the best objective function we have so far (up until T iterations)

- Set the stepsize $\alpha = \frac{1}{\sqrt{T}}$, we have

$$f(x^{\text{best}}) - f(x^*) \leq \frac{1}{2\sqrt{T}}\left(D^2 + G^2\right)$$

Subgradient Method: Convergence

- Differences with the gradient descent
  1. Stepsize $\alpha = \frac{1}{\sqrt{T}}$, have to know how many iterations you are planning to run
  2. The rate is worse, $\mathcal{O}(1/\sqrt{T})$ (compared with gradient descent, $\mathcal{O}(1/T)$), for convex problems
  3. But can deal with nonsmooth function

Subgradient Method: Convergence

- Comparison of rates (running T iterations of all algorithms)

$$\begin{aligned} &\text{second order differentiable, strongly convex} \Rrightarrow && \approx \beta^T \\ &\text{differentiable, convex} \Rrightarrow && \approx (1/T) \\ &\text{nonsmooth, convex} \Rrightarrow && \approx (1/\sqrt{T}) \end{aligned}$$

- **Comment**: If the diminishing rule is chosen, then convergence is guaranteed, α^r is about $1/r$

$$\alpha^r \to 0, \quad \sum_r^{\infty} \alpha^r = \infty, \quad \sum_r^{\infty} (\alpha^r)^2 \leq \infty$$

Outline

1. Motivating Example
2. Subgradients
3. Stochastic Gradient Method (SGD)

SGD: Motivation

- Many machine learning problems minimize empirical loss

$$\min f(\boldsymbol{x}) := \sum_{i=1}^{N} L_i(\boldsymbol{x}; \mathbf{a}_i, b_i)$$

- Computing the gradient

$$\nabla f(\boldsymbol{x}) = \sum_{i=1}^{N} \nabla L_i(\boldsymbol{x}; \mathbf{a}_i, b_i) := \sum_{i=1}^{N} \nabla g_i(\boldsymbol{x})$$

- Per iteration effort needed for computing the gradient is $\mathcal{O}(N)$
- Overall complexity for reaching ϵ-OPT: $\mathcal{O}(\frac{N}{\epsilon})$
- When the number of data points is huge, too bad

SGD: Algorithm

- **Motivation 1**. Reduce the dependence on the # of data points
- **Motivation 2**. Start building models quickly, without seeing all data point once
- **The SGD Method:** $r = 1, \cdots, T$
  1. Randomly pick a data point $i \in \{1, \cdots, N\}$
  2. Compute (sub)gradient $g^r \in \partial L_i(\boldsymbol{x}; \mathbf{a}_i, b_i)$
  3. $\boldsymbol{x}^{r+1} = \boldsymbol{x}^r - \alpha^r g^r$
- Many important variants...

SGD: Benefit

- Very simple and robust
- Starts optimization right away, instead of seeing all (sub)gradients
- Close to incremental methods; but randomness provides some added benefits (e.g., avoiding bad local min)
- Power engine for training modern ML models

SGD: Another Interpretation

- We provide another Interpretation of SGD
- Minimizing a function f
- But each time only an unbiased estimate of its (sub)gradient g^r is available, which satisfies the following

$$\mathbb{E}[g^r] \in \partial f, \; \forall \; r$$

- **The SG method:** $r = 1, \cdots, T$

  1. $\boldsymbol{x}^{r+1} = \boldsymbol{x}^r - \alpha^r g^r$

SGD: Another Interpretation

- For example, suppose there are infinitely many data points $\{\mathbf{a}, b\}$
- The data points follow some unknown distribution $\ell(\mathbf{a}, b)$
- Find a model $\boldsymbol{x}$ minimizes the expected loss over the data set

SGD: Another Interpretation

- Mathematically, we would like to

$$\mathbf{x}^* = \arg\min \ f(\mathbf{x}) = \mathbb{E}_{\mathbf{a},b}[L(\mathbf{x};\mathbf{a},b)] = \int L(\mathbf{x};\mathbf{a},b)\ell(\mathbf{a},b)d\mathbf{a}db$$

- Each time obtain one data point $(\mathbf{a}^r, b^r)$, evaluate subgradient

$$g^r = \partial L(\mathbf{x}, \mathbf{a}^r, b^r)$$

- This is an unbiased estimate of the subgradient of the objective $f(\boldsymbol{x})$, i.e., $\mathbb{E}[\partial L(x, \mathbf{a}^r, b^r)] = \partial f(\boldsymbol{x})$ (why?)

- Perform SGD each time on $\partial L(x, \mathbf{a}^r, b^r)$ (according to the previous page)

SGD: Stepsize and Rate

- The choice of step-sizes: Again assume $\|g^r\| \le G$
  1. Fixed stepsize: $\alpha^r = \frac{\|\boldsymbol{x}^*\|_2}{G\sqrt{T}}$, for all r
  2. Diminishing: $\alpha^r = \frac{\|\boldsymbol{x}^*\|_2}{G\sqrt{r}}$
  3. Unknown G, $\|\boldsymbol{x}^*\|$: $\alpha^r = \frac{\beta\|\boldsymbol{x}^*\|}{G\sqrt{r}}$, for some $\beta > 0$
- The convergence rate:
  1. Fixed stepsize: $\mathbb{E}[f(\bar{x}^r)] - f(\boldsymbol{x}^*) \le \frac{G\|\mathbf{w}^*\|_2}{\sqrt{T}}$
  2. Diminishing: $\mathbb{E}[f(\bar{x}^r)] - f(\boldsymbol{x}^*) \le \frac{4G\|\boldsymbol{x}^*\|_2}{\sqrt{T}}$
  3. Unknown G, $\mathbb{E}[f(\bar{x}^r)] - f(\boldsymbol{x}^*) \le \frac{4G\|\boldsymbol{x}^*\|_2}{\sqrt{T}} \max\{\beta, \frac{1}{\beta}\}$, for some $\beta > 0$
- **Main message**: Rate is in the order of $\mathcal{O}(\frac{1}{\sqrt{T}})$

SGD vs GD vs Subgradient

- Let's compare the rate of SGD and GD, for smooth problem
- SGD Achieves ϵ optimal solution requires $\mathcal{O}(1/\epsilon^2)$ iterations, but each iteration compute 1 gradient.
- GD achieves ϵ optimal solution requires $\mathcal{O}(1/\epsilon)$ iterations, but each iteration compute N gradients (total # of data points).
- When $N = 10^6$, $\epsilon = 10^{-2}$, which one is better?
- For nonsmooth problem, SGD is even better, as both SGD and subgradient method require $\mathcal{O}(1/\epsilon^2)$ iteration to achieve ϵ-OPT

Example: SGD for SVM

- The SVM problem in nonsmooth formulation

$$\min_{\boldsymbol{x}} \quad \frac{1}{N}\sum_{i=1}^{N}\max\{0, 1 - b_i(\mathbf{a}_i^T\boldsymbol{x})\} + \gamma\boldsymbol{x}^T\boldsymbol{x} \tag{0.1}$$

- Recall that one of the subgradient for the hinge loss

$$\partial h(\boldsymbol{x}) = -b_i\mathbf{a}_i, \text{ if } b_i(\mathbf{a}_i^T\boldsymbol{x}) < 1 \text{ (miss-classified)}, \tag{0.2}$$

$$\partial h(\boldsymbol{x}) = 0, \text{ Otherwise} \tag{0.3}$$

- Then the stochastic subgradient method for SVM: for $r = 1, \cdots,$
  1. Pick a data point i
  2. If the data point is miss-classified, do

$$\boldsymbol{x}^{r+1} = \boldsymbol{x}^r - \alpha^r\left(2\gamma\boldsymbol{x}^r - \frac{1}{N}b_i\mathbf{a}_i\right)$$

  3. Else: $\boldsymbol{x}^{r+1} = (1 - \alpha^r 2\gamma)\boldsymbol{x}^r$

SGD: Practical Considerations

- How to select stepsizes?
- Leon Bottou suggests, let $\alpha^r = \frac{\eta}{r}$
- Choose η: Select a small subsample, try various rates $10, 1, 0.1, \cdots$, pick the one that works the best
- How to stop? One way is to create a "validation set" and monitor cost function on the validation set
- The algorithm stops when no decrease of cost evaluated at the "validation set"

More on SVM: Mini-Batch

- Each time takes not 1, not all, but a "mini-batch" of b data points
- Steps:
  1. Update
     $$\boldsymbol{x}^{r+1} = \boldsymbol{x}^r - \frac{\alpha^r}{b} \sum_{i=j}^{j+b-1} \partial L_i(\boldsymbol{x}^r; \mathbf{a}_i, b_i)$$
  2. Move to another batch, $j = j + b$
- Why b data points? When you have good vectorization to parallel computation

Conclusion

- Readings:

1. Boyd's Lecture Notes on Subgradient Methods
2. Ben Rech's Notes on Stochastic OPT